

Nikolaos Genethlis

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Education

University College London — London, UK · *Computer Science MEng (Hons)* · Sep 2022 – Jun 2026 (expected)

- Year 1 grade: 79%, Year 2 grade: 76%, Year 3 grade: 77%
- Dissertation at PhaseCraft on Fermi-Hubbard Circuit Verification
- Academic Merit Scholarship from the government of Cyprus
- Relevant modules: Systems Engineering, Software Engineering, Algorithms & Data Structures, Networked Systems, Security

The English School Nicosia — Nicosia, Cyprus · *High School Diploma* · Sep 2014 – Jun 2021

- Greek Apolytirion 20/20; GCE A* in Further Maths, Maths, and Physics, A in Computer Science and Modern Greek
- Headmaster's award for high achievement, given to the top 1% of students

Technical Skills

Languages: C++, C, Python, TypeScript, JavaScript, Rust, Java, Haskell, Armv64 & x86 assembly

Frameworks & Libraries: React, Redux, Material UI, Node.js, DPDK (Data Plane Development Kit), Pandas, NumPy

Tools: Linux, shell, git, Docker, gdb/lldb, CMake, Vite, Jest, CI/CD

Concepts: Data Structures & Algorithms, Object-Oriented Programming, Concurrency & Multithreading, Unit Testing, Agile

Work Experience

Tild — Software Engineer (Part-time) · *Property Management Platform* · London, UK · Jan 2026 – Present

- Built an in-app AI assistant on Google Gemini whose function-calling tools edit tasks and fetch posts and comments from plain-language requests; users accept or reject each suggested change in one batched step.
- Built a Gantt view for tasks with date fields, filters, and grouping by assignee, and wrote the Jest and React Testing Library tests that cover the new logic.
- Shipped features and bug fixes across the React and TypeScript app (Redux Toolkit, Material UI, Capacitor for web, iOS, and Android) through the team's pull-request and CI/CD workflow.

Squarepoint Capital — C++ Software Engineer Intern · *Market Data, Architecture & Strategic Projects* · London, UK · Jun – Sep 2025

- Enhanced an internal low-latency Network Profiler to support real-time packet analysis, reliably handling throughput of up to 40Gb/s where it had previously read only saved tcpdump and .pcap captures.
- Implemented high-performance kernel bypass using DPDK, achieving a 50% improvement in packet processing speed over the standard Linux networking stack, and kept the capability in-house to save hundreds of thousands of pounds a year against buying or outsourcing an equivalent.
- Developed custom TCP & ICMP filters in C, enabling traffic inspection and filtering within the profiling system.
- Delivered a 30-minute technical presentation and report to 10+ engineers across Market Data and Networking, covering the architecture, metrics, and testing methodology.

Arm — Software Engineer Intern · *Central Technology, CPU Competitive Analysis* · Cambridge, UK · Jun – Sep 2024

- Wrote ARM64 assembly micro-benchmarks, with C and C++ harnesses to set them up, to detect Data Memory-Dependent Prefetchers (DMPs) on 10+ Arm and competitor CPUs, and analysed the results in Pandas and Plotly.
- Traced DMP-related side-channel vulnerabilities from the Augury, Gofetch, and Meltdown/Spectre papers to map the trade-off between performance and security.
- Presented the findings and proposed solutions to CPU designers in a 75-minute Q&A that informed secure-prefetcher design decisions.

Projects & Other Experience

Motion Input — Dysarthric Speech Recognition (Systems Engineering) · *C++, Python* · Sep 2022 – May 2023

- Developed a dysarthric speech-to-text model to make gaming and patient communication more accessible.
- Built a C++ MFC Windows app for the NHS that helped dysarthric patients communicate their needs.

InCite — AI-Powered Citation Mapping for Academic Research (1st Place, IC Hack 2025) · *Python* · Feb 2025

- Developed an AI tool that maps citation networks, flags conflicts of interest, and assesses research methodology.
- Built the responsive React front end and wired in NLP models to interpret citations.

Programming Tutor · *University College London* · Sep 2023 – May 2024

- Ran weekly programming tutorials for 10 first-year students on core programming concepts.
- Selected from 45 candidates on the strength of exam and coursework results.

Cypriot National Guard — Military Service · *Nicosia, Cyprus* · Jul 2021 – Sep 2022

- Liaised with senior UN officers and handled sensitive national-security information with strict attention to detail.